Rotaract Club of Xiamen University Malaysia

Rotary Club of Kuala Lumpur Dynamics

Rotary Club of Tun Razak Exchange

SDC Hackathon

Phase 1 : Preliminary Project Proposal

Hosted by Xiamen University Malaysia (XMUM) AI Club

Submission Deadline: 25 October 2026

Instructions: Complete every field below and submit as a single PDF document named “SDC2026_TeamName_Preliminary.pdf” before the deadline above. Answers must comply with the word limits stated for each open-ended question. Incomplete submissions may not be eligible for review.

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1. TEAM AND PROJECT OVERVIEW

Team Name

MIX MECHS

Selected Domain

Domain 1: The Aiversecity Future-Ready Workforce

Selected SDG Track (1–17)

SDG 4: Quality Education (main, target 4.4); SDG 8: Decent Work and Economic Growth (target 8.5)

Project / Solution Name

SkillFolio

Name (Leader)

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Email & Phone (Leader)

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University Affiliation(s)

University of Southampton Malaysia

Number of Team Members

4

2. EXECUTIVE SUMMARY & PROBLEM STATEMENT

2.1. The Problem Context

Describe the specific, real-world challenge your team is tackling. Who is affected by this problem, and why does it matter? Highlight the critical social, environmental, or economic issues involved.

[Maximum 250 words]

Fresh graduates in Malaysia are starting their careers just as AI changes what employers ask for. In 2024, 32.2% of Malaysian graduates were working in jobs below their qualifications, and the figure was 66.0% for graduates aged 24 and under (DOSM, Graduates Statistics 2024). The Minister of Science, Technology and Innovation said in January 2025 that more than 600,000 workers will need to reskill within three to five years (MOSTI).

Take Aina, 23, a business-analytics graduate in Kuala Lumpur. She has about 200 hours to spend on learning this year and little money. Everyone tells her to learn AI, but she can't see which of her skills employers are asking for more of, which roles she could actually reach, or whether the course she just finished moved her any closer. People switching careers, such as an accountant moving into data, have the same problem.

The tools she can use don't help. A resume is fixed on the day it is written. Career tests give one answer and ignore that AI's effect is still changing. Course sites record that you finished something, not whether it got you nearer a job. Chatbots sound sure but can make up jobs, courses and numbers, and give no sources.

So learning hours get wasted, graduates stay underemployed, and people retrain for skills the market is already moving away from. Choosing what to learn is a bet made under uncertainty, and young people make it with no data and no record of their own progress.

2. EXECUTIVE SUMMARY & PROBLEM STATEMENT

2.2. The Proposed AI Solution

Provide a high-level overview of your proposed application. How does it apply artificial intelligence to solve the problem stated above? Briefly mention the tools or technical skills gained from the SDC Phase 1 workshops (e.g., GenAI APIs, database architecture, cloud hosting) that you intend to leverage.

[Maximum 300 words]

SkillFolio is a web app that turns a resume into a skill portfolio the user keeps up to date as they learn.

The user pastes or uploads a resume. We find the skills in it from a list of 73, and the user sets how well they know each one: Basic, Working or Strong. Keyword matching catches skills that are named. An open AI model (BAAI bge-small, on a RunPod GPU, switched on with two settings) catches skills the resume describes without naming: "built dashboards for managers" suggests Power BI. It can only pick from our list, and the user confirms each suggestion.

Next, the app shows where each skill stands in the job market. Demand comes from 7,961 Hacker News "Who is hiring" posts over 25 months, and AI exposure from the Anthropic Economic Index and O*NET. We estimate each skill's demand 12 months ahead, with a range, under two scenarios: moderate and fast AI progress.

The app then ranks 20 roles by how well the user's skills fit, shows what is missing and which SDGs each role serves, and builds an ordered plan of free courses that raises the fit most per hour of study. As the user marks courses done, their skill levels and fit score update.

Every number shows its source and date, forecasts are labelled as estimates, and the app can only recommend roles, skills and courses from our checked lists. It supports a decision and never makes one. Its limits are stated openly: tech and data roles only, progress is self-reported, and forecasts are estimates.

From the SDC workshops we used a GenAI model (on RunPod), database design (dated data snapshots with tested data contracts) and cloud hosting (Replit).

3. TECHNICAL ARCHITECTURE & IMPLEMENTATION PLAN

3.1. Conceptual Tech Stack

List the technologies, frameworks, APIs, and databases you plan to use to build a functional, deployed prototype. Describe (or attach as an appendix) a simple system block diagram showing how the workshop elements connect into one architecture.

Did your project utilize any official SDC sponsor-provided GenAI API or Cloud Platform? If yes, specify which one and how it is used.

Front end: Streamlit 1.66 (Python) multipage web app, Plotly charts, works on phone and laptop

Engine: Python 3.12, pandas, NumPy: skill forecasts (log-linear trend, shrinkage, scenario shift), portfolio metrics, role matching, course planner, SDG allocation

AI model: BAAI/bge-small-en-v1.5 embedding model (sentence-transformers, PyTorch) on a RunPod Serverless GPU endpoint: resume-to-skills matching

Data pipeline: Python + Requests: HN Algolia API → skill extraction → dated JSON snapshots; Anthropic Economic Index and O*NET joined per skill

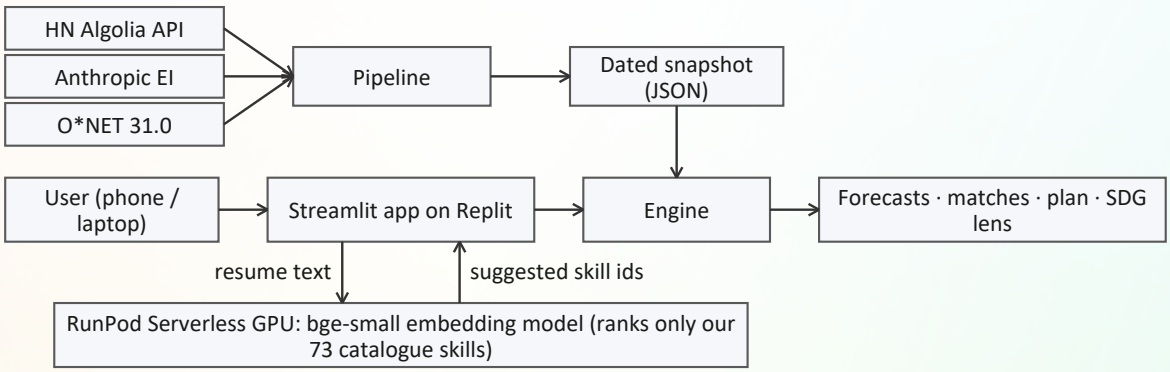
Data store: Versioned CSV and JSON files in Git (skills, roles, courses, profiles, snapshots), each checked by automated tests against a written data contract

Progress tracking: Course progress (0/25/50/75/100%) set by the user and kept in the browser session; finishing a course raises the skill level in the profile and re-runs the engine

Hosting: Replit (development and public deployment), RunPod (GPU model)

Quality: pytest suite (over 900 tests), GitHub pull requests

System block diagram



Sponsor-provided GenAI API or cloud platform? Yes. **RunPod** (sponsor credits): a Serverless GPU endpoint runs the embedding model that turns resume text into suggested skills. **Replit** (hackathon Replit Core): development workspace and public hosting of the live prototype.

3. TECHNICAL ARCHITECTURE & IMPLEMENTATION PLAN

3.2. Design-Thinking & Deployment Strategy

Explain how your team applied design-thinking methodology to ensure this solution is technically viable, user-friendly, and ready for real deployment by Phase 2.

[Maximum 200 words]

We began with one person and one decision: a graduate with about 200 hours choosing what to learn. DOSM and MOSTI data confirmed the problem. We wrote five made-up personas, from an accounts clerk to an engineering graduate aiming at renewable energy, and tested the app against them.

We compared six ideas against the judging criteria and picked the skill portfolio, because people already understand holding something and rebalancing it.

Before building screens we agreed the data formats and wrote tests, so four people could work at once. The tests enforce our main rules: no invented courses, every number has a source and date, and no age or gender in profiles.

We ran every persona through the engine, tuned the AI model on made-up resumes, and checked the app on phone and laptop.

The app runs on Replit from our public GitHub repository. The AI model runs on RunPod when switched on, and our public demo runs in keyword mode so nobody waits for a cold start. A saved data snapshot covers a data source going down. Phase 2 plans saved portfolios, Malaysian job data for other industries, and Bahasa Malaysia.

4. PRELIMINARY ENVIRONMENTAL & SDG IMPACT FRAMEWORK

To align with SDC's focus on meaningful global sustainability, establish your project's impact baseline before building the prototype. Use estimated data, published research, or public benchmarks where exact figures are unavailable. Provide the citation of any sources used.

The target, baseline, and KPI you provide here will be used by judges during shortlisting to assess Innovation and Creativity as well as Feasibility of build. Vague or unsupported projections may weaken your case for advancing to Phase 2.

4. PRELIMINARY ENVIRONMENTAL & SDG IMPACT FRAMEWORK

Metric Category	Your Response
Target SDG Sub-Target(s) *Identify the exact UN SDG sub-target your project addresses* *e.g., Target 11.6 "Reduce the adverse per-capita environmental impact of cities, focusing on air quality and waste management."*	**Target 4.4:** "By 2030, substantially increase the number of youth and adults who have relevant skills, including technical and vocational skills, for employment, decent jobs and entrepreneurship." Supporting: **Target 8.5:** "By 2030, achieve full and productive employment and decent work for all women and men, including for young people…"
Current Problem Baseline *What is the starting condition of the problem, expressed as a number?* *e.g., "The target infrastructure currently experiences 35% energy inefficiency due to unoptimized scheduling."*	66.0% of Malaysian graduates aged 24 and under worked below their qualifications in 2024 (overall graduate rate 32.2%, 1.60 million people) (DOSM, Graduates Statistics 2024). More than 600,000 workers must reskill within 3–5 years (MOSTI, Jan 2025). A graduate today has no tool that ties their learning hours to dated demand data: 0% of the recommendations they receive are backed by a cited, dated source.

4. PRELIMINARY ENVIRONMENTAL & SDG IMPACT FRAMEWORK

Metric Category	Your Response
Core Measurable KPI *The specific unit you will track* *e.g., % reduction in power consumption, processing-latency improvement, or kg of resource waste avoided*	(1) Target-role fit % before vs after the recommended plan, within the user's hour budget. (2) Portfolio AI-exposure score before vs after the plan. (3) Share of recommended roles, courses and numbers with a cited, dated source. (4) Plan completed (hours done / planned hours) and the fit change it produces. (5) Time from profile to plan, and whether a test user can name a reachable role and their next course afterwards.
Phase 2 Impact Target *Your realistic, percentage-based target by the final prototype presentation* *e.g., "20% improvement in computational resource use."*	Across our 5 synthetic personas: ≥ 20% average increase in target-role fit % and ≥ 10% average reduction in portfolio AI exposure within each persona's hour budget; 100% of recommended roles, courses and numbers cited and dated; profile to plan in under 3 minutes, with ≥ 80% of 10 outsider test users able to name a reachable role and their next course.

Sources

• DOSM, Graduates Statistics 2024: https://www.dosm.gov.my/uploads/release-content/file_20260401145140.pdf

• Malay Mail, "Mosti: AI disruption looms in Malaysia, 600,000 workers must reskill in three to five years", 7 Jan 2025: https://www.malaymail.com/news/malaysia/2025/01/07/mosti-ai-disruption-looms-in-malaysia-600000-workers-must-reskill-in-three-to-five-years/162368

• UN Statistics Division, SDG targets 4.4 and 8.5: https://unstats.un.org/sdgs/metadata/?Text=&Goal=4&Target=4.4 · https://unstats.un.org/sdgs/metadata/?Text=&Goal=8&Target=8.5

• Massenkoff & McCrory, Labor market impacts of AI: A new measure and early evidence, Anthropic, 5 Mar 2026: https://www.anthropic.com/research/labor-market-impacts

5. ELIGIBILITY & READINESS CHECKLIST

Tick all boxes that apply before submitting.

- ☑ All team members have completed the mandatory GenAI, Database, and Cloud Deployment upskilling workshops.
- ☑ The project concept addresses a verified real-world problem mapped directly to one or more UN SDGs.
- ☑ The proposed solution targets a functional, deployed application rather than a theoretical business case study.
- ☑ All fields in this template have been completed within the word limits stated.

6. IP & ORIGINALITY DECLARATION

6.1. Pre-Existing / Prior-Owned Work

Does this submission include any code, design, research, or other work that any team member created or owned before the start of this hackathon?

☐ Yes ☑ No

If yes, briefly describe what was pre-existing and which team member(s) owned it:

6. IP & ORIGINALITY DECLARATION

6.2. Third-Party Code, Datasets, or Models

Does your project use any third-party code libraries, open-source frameworks, public/private datasets, or pre-trained AI models that your team did not create?

☑ Yes ☐ No

If yes, list them below (open-source use is permitted provided it is declared and properly licensed/attributed):

Name	Type (code/dataset/model)	License
Streamlit 1.66 pandas 3.0 NumPy 2.5 Plotly 7.1	code code code code	Apache-2.0 BSD-3-Clause BSD-3-Clause MIT
Requests 2.34 pypdf 6.21 sentence-transformers PyTorch RunPod Python SDK pytest 9.1	code code code code code code (testing)	Apache-2.0 BSD-3-Clause Apache-2.0 BSD-3-Clause MIT MIT
BAAI/bge-small-en-v1.5 Hacker News "Who is hiring" posts via HN Algolia API	model (pre-trained) dataset	MIT Public API
Anthropic Economic Index, job exposure O*NET 31.0 Technology Skills UN SDG targets	dataset dataset dataset	CC-BY CC BY 4.0 UN public data
Course list (52 free courses: freeCodeCamp, Kaggle, Microsoft Learn, Coursera audit, Cisco NetAcad and others)	links and metadata only	Each provider's public listing

6. IP & ORIGINALITY DECLARATION

6.3. Employment-Related IP Conflicts

Is any team member currently bound by an employment, internship, or consulting agreement under which their employer could claim ownership of work they create during this hackathon?

☐ Yes ☑ No

If yes, please describe the nature of the agreement and which member it concerns:

7. DECLARATION

By submitting this proposal, the team leader confirms that the information provided is accurate to the best of the team's knowledge and that the project will be original work created during the SDC 2026 hackathon period.

[Signature]

[Date]

25/10/26